Circle: Formula Sheet

Equation and centre

Centre-radius: $(x-h)^2+(y-k)^2=r^2$

General: $x^2+y^2+2gx+2fy+c=0$

Centre, radius: $(-g,-f)$, $\sqrt{g^2+f^2-c}$

Circle condition: $a=b\neq 0$, $h=0$

Nature: $g^2+f^2-c>0$, $=0$, <0

Diametric: $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

Parametric: $(h+r\cos\theta,\ k+r\sin\theta)$

Chords and intercepts

Chord: $2\sqrt{r^2-p^2}=2r\sin\theta$

Intercepts: $2\sqrt{g^2-c}$, $2\sqrt{f^2-c}$

Axis: $g^2>c$ cuts, $=c$ touches, $<c$ misses

Any curve: $|x_1-x_2|\sqrt{1+m^2}$

Given mid-point: $T=S_1$, length $2\sqrt{-S_1}$

Chord $P(\alpha)Q(\beta)$:

$$x\cos\frac{\alpha+\beta}{2}+y\sin\frac{\alpha+\beta}{2}=a\cos\frac{\alpha-\beta}{2}$$

Point and line

Position: $S_1>0$ out, $=0$ on, <0 in

Distances: $AC\pm r$; line: $p-r$

Tangency: $p=r$; $c^2=a^2(1+m^2)$

Line types: $p>r$, $p=r$, $p<r$, $p=0$

Tangents

At a point: $T=0$

Parametric: $x\cos\theta+y\sin\theta=a$

Slope form: $y=mx\pm a\sqrt{1+m^2}$

Tangents at α,β meet:

$$\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}},\frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$$

Length: $L=\sqrt{S_1}$; power $S_1=PA\cdot PB$

Pair: $SS_1=T^2$; chord of contact: $T=0$

Quadrilateral $PACB=RL$; $AB=\frac{2RL}{\sqrt{R^2+L^2}}$

$\triangle PAB=\frac{RL^3}{R^2+L^2}$; angle $2\tan^{-1}\frac{R}{L}$

Circumcircle of PAB: $(x-x_1)(x+g)+(y-y_1)(y+f)=0$

Director circle: $x^2+y^2=2a^2$

Pole of $lx+my=n$:

$$\left(\frac{a^2l}{n},\frac{a^2m}{n}\right)$$

Family of circles

Two circles: $S_1+\lambda S_2=0$

Circle and line: $S+\lambda L=0$

Two points:

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)+\lambda L_{AB}=0$$

Touching L at P: $(x-x_1)^2+(y-y_1)^2+\lambda L=0$

Triangle:

$$L_1L_2+\lambda L_2L_3+\mu L_3L_1=0$$

Quadrilateral: $L_1L_3+\lambda L_2L_4=0$

Two circles

DCT: $\sqrt{d^2-(r_1-r_2)^2}$; TCT: $\sqrt{d^2-(r_1+r_2)^2}$

Meet of DCTs / TCTs: $r_1:r_2$ external / internal

Angle of tangents: $\sin\theta=\frac{r_1\mp r_2}{d}$

Separate $d>r_1+r_2$: 4 tangents

External touch $d=r_1+r_2$: 3

Intersect $|r_1-r_2|<d<r_1+r_2$: 2

Internal touch $d=|r_1-r_2|$: 1; inside: 0

Radical axis and orthogonality

Radical axis / common chord: $S_1-S_2=0$

Bisecting circumference of S_1: C_1 on $S_1-S_2=0$

Coaxial: $S_1+\lambda(S_1-S_2)=0$

Line-circle angle: $\cos\theta=\frac{p}{r}$

Circle-circle angle:

$$\cos\theta=\frac{r_1^2+r_2^2-d^2}{2r_1r_2}$$

Orthogonal: $2g_1g_2+2f_1f_2=c_1+c_2$, $r_1^2+r_2^2=d^2$

Common chord (orthogonal): $\frac{2r_1r_2}{d}$